SWEEP rule on the KD-Toric code

March 15, 2026

1 Investigation

[COMMENT] To do: understands how the investigation should look like. Plot an example to such tree. The model goes as follow, at each even iteration we toss a noise, then in each odd iteration we perform a single SWEEP rule iteration. That allows to track over the history of the errors.

We say that a check $e \in$ Error, if there is a face f that adjoin to e and belongs to the Error.

Definition 1.1 (Investigation)**.** For any unsatisfied check e at time t, we define V (the set of checks which participated in 'misleading' of e), and two sets of edges on V, Arrows and Forks via the following algorithm:

let $V = \{e\}$, Arrows $= \emptyset$, Forks $= \emptyset$
let $Q \leftarrow$ queue $\{e\}$
while *Q is not empty* **do**
    $e' \leftarrow$ dequeue Q
    if *$e' \notin$ Error* **then**
        $e_1, .., e_l \leftarrow$ Excuse(e')
        Insert $\{e_i, e'\}$ to Arrows
        Insert $\{e_i, e_j\}$ to Forks
    end
end

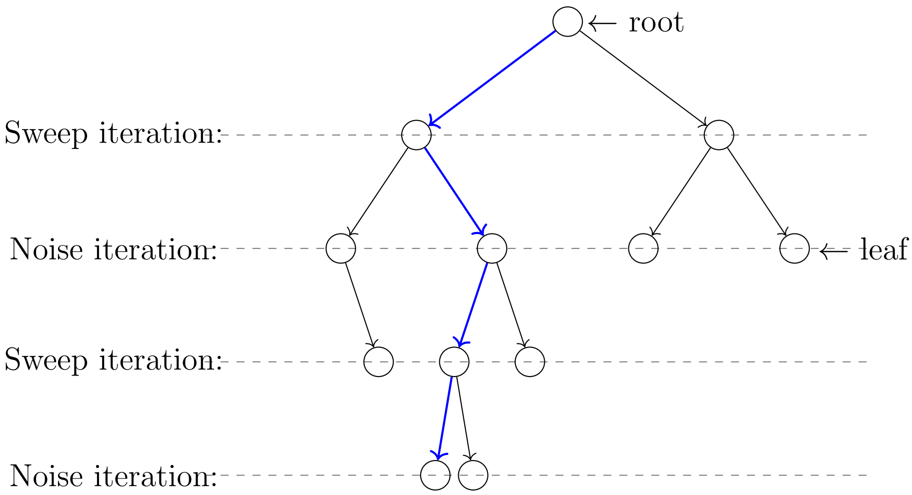

Figure 1: blabla